

Interactive writing systems and why small(er) could be more beautiful

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ABSTRACT

Machine learning models can support human creativity, including tasks such as writing. This position paper explores and critiques the design of writing systems based on big data. The paper proposes small data as a way to think about systems for underrepresented languages, types of writers, and writing tasks. In addressing these topics, the paper also considers how we can make writing assistants that are more accessible and inclusive than the state-of-the-art.

CCS CONCEPTS

• **Computing methodologies** → **Natural language generation**; **Unsupervised learning**; • **Human-centered computing** → HCI theory, concepts and models.

KEYWORDS

writing assistants, data sets, large language models, co-creativity

1 INTRODUCTION

Karimi et al suggest three ways to characterise the role of humans in creative systems: fully autonomous systems (e.g GPT-3), creativity support tools (e.g a rhyming dictionary), and co-creative systems (e.g Verse by Verse) [12, 22, 34]. Interactive writing assistants cover all three types, though the paper focuses on fully autonomous and co-creative systems. Fully autonomous Large Language Models (LLMs) are now capable of generating texts that approach or surpasses that of *some* humans [4, 5, 11, 27]. State-of-the-art LLMs are able to perform well across language domains such as question answering or summarizing. Essentially, LLMs predict the next word, or series of words, in a text sequence. The models are trained large datasets of words, or *tokens* (pieces of words) such as the Pile or Common Crawl. The datasets range from hundreds of GiB to TiB [3, 9, 11]. Research suggests there is strong correlation between model size and performance [18]. Despite recent achievements, LLM limitations include (a) expense and difficulty performing inference on; (b) model decisions are not easily interpretable; (c) disadvantaged social groups have been excluded from their development and, (d) most LLMs are trained on English-language text that contains data biases [2, 5, 28]. Typically, LLMs also reflect biases and overtly abusive language patterns in training data. This can lead to harms for some users [2]. Although there are studies exploring bias mitigation through computational techniques, in their study describing GPT-3 the authors caution against over reliance on computational solutions [5, 13, 30]. By virtue of their size, LLMs models face issues around availability of training data, sociocultural bias, and environmental costs [2, 28]. To put the costs into context, it is estimated to cost £ millions to train LLMs [29]. This results in most LLMs being developed and owned by private technology companies as shown in table 1. A potential way to mitigate against these issues, pertinent to creative arts, is to work with smaller-scale

data sets. As Viglienconi et al argue, "working with small-scale data sets is an often-overlooked but powerful mechanism for enabling greater human influence over generative AI systems in creative contexts." [33]. Smaller data sets could also reduce concerns about the way LLMs might constrain rather than augment human capacity [1, 19, 23]. To give a sense of proportion for the term *large* with reference to language models, table 1 contains a summary of notable state of the art models.

2 CO-CREATIVE SYSTEMS AND SCALE

Current work on co-creative settings focuses on the role of the system as a generator augmenting what people can achieve in creative tasks [1]. Some problems with this approach include; aligning system capabilities and user expectations, language model bias, system interpretability, and user interaction design [6, 19, 35]. For instance, studies have found that the mental expectation of users affects their strategies and perception of the system's role in the co-writing process [1, 35]; too much automated creation can be at the expense of human users [1, 6]. As Calderwood et al point out, writers are not looking for language models to solve NLP tasks but rather "writers engaged with co-creative systems are looking for creative insight" [7]. Csikszentmihalyi's study on creativity supports this point in finding that the goal of a writer is to develop an individual 'voice'. [10]. Given the diversity in writers' goals and competencies, a well-designed co-creative system should cater for multiple support needs [1, 6, 19]. LLMs such as GPT-3 at present lack explicit support for the user in terms of explainability of their outcomes (i.e generated text), or references and/or influences for generated text output.

3 POETRY AND SMALL(ER) DATA

A study by Sheng and Uthus states that "...creative works are often shaped by the lived experiences and timely issues of the creator's life..." [30]. Designing writing assistants for underrepresented languages, types of writers, and/or writing tasks requires a departure from current big data LLMs. Instead of large data-driven LLMs, working with small-scale data can be beneficial in terms of accessibility, underrepresented languages, types of writers and/or writing tasks [32, 33]. As an example, the ReRites project involved fine-tuning an open source LLM (GPT-2) on the artist's custom poetry corpus to generate poems in real-time [16]. Another GPT-2 experiment fine-tuned the model with a poetry corpus [24]. While the results of the GPT-2 experiments were not state-of-the-art, they generated specialized outputs based on small rather than big data. Arguably, the key challenge both experiments have is their dependency on GPT-2. An alternative (hypothetical) approach is to leverage work within machine learning and bias. Sociocultural bias has gathered research attention [2, 13, 17]. Considering bias in design terms offers possibilities of systems built on underrepresented

Language Model Characteristics

Table 1: Summary of State-of-the-Art Language Models by Size, Model Type and Ownership

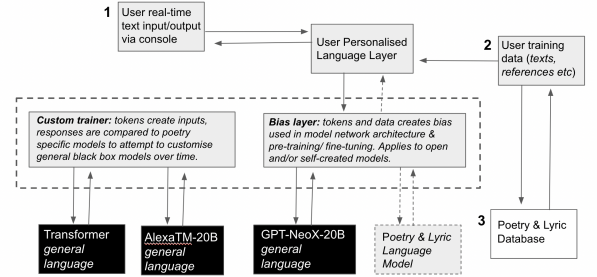
Model Name	Parameters	Model Type	Owner
LaMDA	137 billion	Transformer	Google
GPT-3	175 billion	Transformer	OpenAI
ChatGPT/InstructGPT	175 billion	Transformer	OpenAI
BLOOM	176 billion	Transformer	BLOOM Project
Megatron-Turing NLG	530 billion	Transformer	Microsoft and NVIDIA
PaLM	540 billion	Transformer	Google
GLaM	1 trillion	Mixture of Experts	Google

sociocultural resources. In the case of poetry writing, including song lyrics as a central part of the training dataset could enhance writing assistants through (a) data reflecting popular contemporary issues; and (b) bringing voices from excluded communities into NLP conversations. Rather than designing LLMs, this approach considers multiple types of *Small or Specialized Language Models (SLMs)*. [21] *Bias* in this context plays a key technical and social and cultural role. Vigliensoni et al propose the idea that small, manually-curated datasets may be desirable as a way for creators to communicate to co-creative systems sorts of outputs they would like to generate [33]. They go on to assert that "when datasets are small, changes such as the addition or removal of a few training examples can also have appreciable effects on the trained model's behaviour." [33] Potentially, the benefits of SLMs are writing systems that are lower cost, and more suitable for experimentation by diverse communities. As an indicator of scale when referring to *small* and *large* datasets it is instructive to use the example of (hypothetical) poetry writing system. The following sections explore this idea in more detail

Poetry and Small Data: LLMs are trained on a diverse source of documents such as web pages, articles, books, code repositories and other data [3, 11]. Typically, the datasets contain hundreds of millions to billions of documents. Examples include the Pile (211m documents) and Common Crawl (3.5 billion) [3, 9]. In contrast, a poetry system could include all the most *critically important* poetry and song lyrics in the English language and still be around 10 percent or less of the size of a standard LLM training dataset. To illustrate the numbers, an using GPT-3 as an example. The GPT-3 model was trained on the Common Crawl dataset, representing nearly one trillion words from diverse sources [5]. The English language poetic cannon, works considered to be central to the identity of the literary tradition or culture, consists of around 342,000 poems [25, 26]. In the case of song lyrics, the major record companies control the copyright to around 10m songs [15]. Assuming a combination of this data represented the *best* text in terms of quality, it would be less than 10.5m documents as against a standard LLM figure of 200m plus documents. It is of course arguable that the dataset should include more contemporary poems, lyrics not controlled by record companies, and non-English text. Even so, it is still likely to be far smaller than LLM datasets - most of which are also limited to English language [28].

Continuing with the use case of poetry writing, Olatunji suggests an approach that tries to leverage LLMs as part of SLM. The objective for the model in this instance is to leverage the benefits of LLMs to create smaller, more personal models, specific to the writing style of each user [21]. The approach is illustrated below (figure 1).

Figure 1: Sparse and Dense Network Model Elements



(1) Text input by user is returned as partially completed poetic text and/or poetic and lyrical recommendations for the user to consider. (2) User personalised data submitted as poems or lyrics and/or recommendations of favourite artists and their work. These are used to create a corpus of user text. Prior examples of user generated text uploaded to system; recommender and/or database search to enhance user text with additional poetic texts (e.g from web crawl) (3) Database of poetic texts as well as metadata such as rhyme scheme and Parts of Speech (PoS).

The system design supports multiple *states* that from full automation, to prompts acting as a starting point for the user. The goal of the system is to augment the capacity of human users to create poetry; this could be external to the system. An example of this is where the system uses the user's text as a creative catalyst by to recommend other (related) texts for the user to read and reflect upon. Part of a writers development will be a result of the poetry they have previously been exposed to. Robert Graves stated that, "only a poet of experience...can hope to put himself in the shoes of his predecessors, or contemporaries, and judge their poems by recreating technical or emotional dilemmas which they faced while at work on them." The argument is that experience of (reading) poetry is a prerequisite to writing poetry; in other words, to become

a writer it is necessary to be a reader. The implication for a computational system is that a writing assistant might support its user as much by making *intelligent* recommendations as by generating text on the the user's behalf.

3.1 Small Data and Inclusive Design

Emerging from African American youth in the 1970s, rap (as part of hip-hop culture) is one of the most popular music genres in the world [8, 14, 20]. It is listened to by 1.85 billion people, or 26 percent of global music listeners [31]. However, rap is usually not included as an important part of discussions on co-creative systems and poetry writing. The benefits of using extensive rap lyrics within LM data sets include:

- Training data that represents wider audience concerns, thoughts and feelings.
- Training data will be dynamic and reflect contemporary sociopolitical issues.
- Opens up the possibility of bring voices from excluded communities into the NLP community.
- LMs would be enhanced by a linguistically rich and varied source of data..
- Allows lyrics to be part of a wider conversation which potentially generates. new research insight (for computational, language and social researchers).
- The relatively small size of datasets (e.g subset of all lyrics controlled by record companies) would allow systems to use *all* text from the genre.

The last point is perhaps the most important in computational terms. Unlike LLMs which cannot accurately represent languages and creative text (due to training constraints for example) it would be possible (in theory) to build a poetry co-creative system that included *the entire English cannon* and *all of hip hop* from its birth to the present day. The system would be far smaller but also extensive (linguistically) than current LLMs. It would also be cheaper and have lower running costs. With regards to usage, it is conceivable that by using popular music, the system might offer a way to engage underrepresented groups into the HCAI community.

4 CONCLUSION

The paper argues that writing assistants that are dependent on LLMs are potentially diminishing human creativity at a societal level. They do this through limitations within training data as well as their high development costs. As a counter to big data, smaller and more specialized datasets are proposed. These datasets could form the basis for newer types of writing assistant which are highly personalized for, culturally flexible and potentially more responsive towards each individual user [21].

ACKNOWLEDGMENTS

This work research was supported by the Engineering and Physical Sciences Research Council. The author would also like to acknowledge the invaluable support of colleagues at Swansea University;

Professor Matt Jones, Dr Alma Rahat, and Dr Amanda Rodgers. Additional thanks to Tracey McNulty and Swansea Council for their encouragement.

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